Synoptic Question 11: The Physics of a Formula 1 Racing Car

Reading Passage

Read the following article and then answer the questions that follow.

A Formula 1 (F1) car is an extreme example of automotive engineering, designed to achieve the fastest possible lap times. This requires not only a powerful engine but also incredibly high cornering speeds and braking capabilities.

A modern F1 car uses a hybrid V6 turbo engine, which combines a traditional internal combustion engine (ICE) with an Energy Recovery System (ERS). The ERS includes a Motor Generator Unit-Kinetic (MGU-K), which acts as a generator during braking. It converts some of the car's kinetic energy, which would otherwise be lost as heat in the brakes, into electrical energy. This energy is stored in a high-voltage battery. During acceleration, the MGU-K reverses its function, acting as a motor to deliver additional power to the wheels.

Cornering performance is dominated by aerodynamics. The car's wings and bodywork are designed to create a powerful downward aerodynamic force, known as "downforce". This force, which increases with the square of the car's speed, presses the car onto the track, dramatically increasing the normal force on the tyres. This allows for a much greater frictional force between the tyres and the track, enabling the car to take corners at speeds that would be impossible for a conventional vehicle.

The sound of an F1 car is also distinctive. As the car passes a spectator at high speed, the observed pitch of the engine sound changes dramatically due to the Doppler effect.

Data:

Data for a typical F1 car:

Table 1: Data for Formula 1 Car

Parameter	**Value**
Total mass of car and driver, m	800 kg
Time for 0 to 100 km/h acceleration	2.6 s
Maximum power of internal combustion engine (ICE)	750 kW
Maximum speed on a long straight, v_{top}	350 km/h
Speed during a high-speed corner, v_c	180 km/h
Radius of this corner, r	80 m
Aerodynamic downforce at 180 km/h, F_{down}	15000 N
Coefficient of static friction for tyres, μ_s	1.5
Braking speed range for MGU-K	300 km/h to 100 km/h
Time taken for this braking, t_{brake}	2.0 s
Efficiency of MGU-K energy recovery, η_K	60%
True frequency of engine sound, f_s	800 Hz
Speed of sound in air, v_{sound}	343 m s^{-1}

Questions

(a) During the start of a race, the car accelerates from 0 to 100 km/h in 2.6 s.

  (i) Convert 100 km/h to m s^{-1}.

  (ii) Assuming constant acceleration, calculate this acceleration.

  (iii) Calculate the average resultant force on the car during this acceleration.

(b) When the car is travelling at its top speed of 350 km/h, the engine is producing its maximum power of 750 kW. Calculate the total resistive force (aerodynamic drag and friction) acting on the car at this speed.

(c) The car negotiates a flat, curved section of the track with a radius of 80 m at a constant speed of 180 km/h.

  (i) Calculate the centripetal force required to keep the car on this circular path.

(ii) Calculate the total downward normal force exerted by the track on the tyres.

(iii) Calculate the maximum possible frictional force between the tyres and the track.

(iv) By comparing your answers, determine whether it is possible for the car to make this turn.

(d) The MGU-K is used during a braking event from 300 km/h to 100 km/h.

(i) Calculate the total loss in kinetic energy of the car during this event.

(ii) Calculate the amount of electrical energy (in joules) recovered and stored in the battery.

(iii) Calculate the average electrical power generated by the MGU-K during this braking time.

(e) A spectator stands by the track. The car approaches and passes them at 300 km/h.

(i) Calculate the frequency of the sound the spectator hears as the car is approaching.

(ii) Calculate the frequency the spectator hears as the car is moving away.

Solutions

(a) (i) $v = 100 \text{ km/h} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 27.78 \text{ m s}^{-1}$.

(ii) $a = \frac{v-u}{t} = \frac{27.78-0}{2.6} = 10.68 \text{ m s}^{-2} \approx 10.7 \text{ m s}^{-2}$.

(iii) $F_{net} = ma = (800)(10.68) = 8544 \text{ N} \approx 8540 \text{ N}$.

(b) $v_{top} = 350 \text{ km/h} \times \frac{1000}{3600} = 97.22 \text{ m s}^{-1}$.
At constant top speed, the engine's thrust force equals the resistive force.
Using $P = Fv$:
$F_{resist} = \frac{P_{max}}{v_{top}} = \frac{750\times10^3 \text{ W}}{97.22 \text{ m s}^{-1}} = 7715 \text{ N} \approx 7720 \text{ N}$.

(c) (i) $v_c = 180 \text{ km/h} \times \frac{1000}{3600} = 50.0 \text{ m s}^{-1}$.
$F_c = \frac{mv^2}{r} = \frac{(800)(50.0)^2}{80} = 25000 \text{ N}$.

(ii) The total normal force N is the sum of the car's weight and the aerodynamic downforce.
$W = mg = (800)(9.81) = 7848 \text{ N}$.
$N = W + F_{down} = 7848 + 15000 = 22848 \text{ N}$.

(iii) $F_{fric,max} = \mu_s N = (1.5)(22848) = 34272 \text{ N}$.

(iv) The required centripetal force is 25000 N. The maximum available frictional force is 34272 N. Since $F_{fric,max} > F_c$, the car can successfully make the turn.

(d) (i) $v_1 = 300 \text{ km/h} = 83.33 \text{ m s}^{-1}$. $v_2 = 100 \text{ km/h} = 27.78 \text{ m s}^{-1}$.
$\Delta KE = \frac{1}{2}mv_1^2 - \frac{1}{2}mv_2^2 = \frac{1}{2}(800)(83.33^2 - 27.78^2)$
$\Delta KE = 400(6944 - 771.7) = 2.469 \times 10^6 \text{ J} \approx 2.47 \times 10^6 \text{ J}$.

(ii) $E_{recov} = \Delta KE \times \eta_K = (2.469 \times 10^6) \times 0.60 = 1.48 \times 10^6 \text{ J}$.

(iii) $P_{charge} = \frac{E_{recov}}{t_{brake}} = \frac{1.48\times10^6 \text{ J}}{2.0 \text{ s}} = 740000 \text{ W} = 740 \text{ kW}$.

(e) $v_{source} = 300 \text{ km/h} = 83.33 \text{ m s}^{-1}$.

(i) Approaching: $f' = f_s \left(\frac{v_{sound}}{v_{sound}-v_{source}}\right)$
$f' = 800 \left(\frac{343}{343-83.33}\right) = 800 \left(\frac{343}{259.67}\right) = 1057 \text{ Hz} \approx 1060 \text{ Hz}$.

(ii) Moving away: $f' = f_s \left(\frac{v_{sound}}{v_{sound}+v_{source}}\right)$
$f' = 800 \left(\frac{343}{343+83.33}\right) = 800 \left(\frac{343}{426.33}\right) = 644 \text{ Hz}$.

[Diagram of F1 car in a turn, showing $\text{F}_c, F_down, W, andF_friction$]

Figure 1: Forces on an F1 Car During Cornering

Cambridge International AS & A Level Physics (9702)

Paper 4: Structured Questions

Variant 2 (Higher Difficulty) — F1 Racing Car Context

Total for this section: 40 marks

Comprehension passage (read before answering)

During a qualifying simulation, an F1 car and its systems were logged over one representative lap. The total mass of car plus driver was recorded as 7.98×10^2 kg. Maximum usable engine power on the straight was 7.20×10^5 W, while aerodynamic drag in straight-line running was modelled by $F_r = kv^2$ with $k = 0.34\,\mathrm{N\ s^2 m^{-2}}$. At the studied corner, radius was 1.35×10^2 m, tyre-road coefficient was $\mu = 1.65$, and aerodynamic downforce followed $D = cv^2$ with $c = 2.20\,\mathrm{N\ s^2 m^{-2}}$. In a heavy braking phase, speed reduced from $9.2 \times 10^1\,\mathrm{m\ s^{-1}}$ to $2.8 \times 10^1\,\mathrm{m\ s^{-1}}$ over 2.6 s. Data analysis estimated that 80% of lost kinetic energy entered four identical brake discs, each of mass 1.20 kg and specific heat capacity $4.60 \times 10^2\,\mathrm{J\ kg^{-1} K^{-1}}$. Take $g = 9.81\,\mathrm{m\ s^{-2}}$.

Question 1: Dynamics and energy of an F1 car [14]

An F1 car of mass m accelerates from rest along a straight track. Its engine delivers constant power P. Ignore rolling resistance initially.

(a) Show that acceleration a can be expressed as

$$a = \frac{P}{mv}.$$

[2]

(b) Derive an expression for speed v as a function of time t, and show that

$$v = \sqrt{\frac{2Pt}{m}}.$$

[4]

(c) Using values from the comprehension passage, calculate the time taken for the car to reach $v = 8.0 \times 10^1\,\mathrm{m\ s^{-1}}$. [2]

(d) At high speed, resistive force is modelled by $F_r = kv^2$. Determine the terminal speed when engine power remains P. [3]

(e) Open-ended: explain why a constant-power model gives a better high-speed description than a constant-force model for an F1 car. Include at least one quantitative and one physical argument. [3]

Question 2: Cornering, downforce and friction limits [13]

During a flat corner of radius r, tyre-road coefficient of friction is μ. At this speed, aerodynamic downforce is $D = cv^2$.

(a) Show that the maximum available lateral friction force is

$$F_{\max} = \mu(mg + D).$$

[2]

(b) By equating centripetal force requirement to $F_{\max}$, derive an expression for maximum cornering speed $v_{\max}$. [4]

(c) Using data from the comprehension passage, calculate $v_{\max}$. [3]

(d) Open-ended: propose one realistic engineering change to increase cornering speed by at least 5% without changing engine power. Justify using physics. [4]

Question 3: Braking, thermal physics and measurement design [13]

An F1 car enters a braking zone at speed u and exits at speed v. Use the comprehension passage for all numerical values in this question.

(a) Calculate the total loss in kinetic energy of the car. [3]

(b) Determine the temperature rise of one disc for a single heavy braking event. [4]

(c) If the braking event lasts 2.6 s, estimate the average braking power dissipated by all brakes. [2]

(d) Open-ended: describe two limitations of the model above and explain how each limitation would affect the predicted disc temperature. [4]

Sample solution outline (Variant 2)

Q1

(a) $P = Fv$ and $F = ma \Rightarrow P = mav \Rightarrow a = P/(mv)$.

(b)

$$\frac{dv}{dt} = \frac{P}{mv} \Rightarrow v\,dv = \frac{P}{m}dt \Rightarrow \frac{1}{2}v^2 = \frac{P}{m}t \Rightarrow v = \sqrt{\frac{2Pt}{m}}.$$

(c)

$$t = \frac{mv^2}{2P} = \frac{(7.98 \times 10^2)(80)^2}{2(7.20 \times 10^5)} \approx 3.55\,\text{s}.$$

(d) At terminal speed, $P = F_r v = kv^3 \Rightarrow v_t = (P/k)^{1/3} \approx 1.28 \times 10^2\,\text{m s}^{-1}$.

(e) Constant force predicts constant acceleration and unbounded power demand at high v. Constant power predicts decreasing acceleration with speed, matching real drivetrain and drag-limited behaviour.

Q2

(a) Normal load $N = mg + D$, so limiting friction $F_{\max} = \mu N = \mu(mg + D)$.

(b)

$$\frac{mv^2}{r} = \mu(mg + cv^2) \Rightarrow v^2\left(\frac{m}{r} - \mu c\right) = \mu mg \Rightarrow v_{\max} = \sqrt{\frac{\mu mg}{\frac{m}{r} - \mu c}}.$$

(c)

$$v_{\max} = \sqrt{\frac{(1.65)(7.98 \times 10^2)(9.81)}{\frac{7.98\times10^2}{1.35\times10^2} - (1.65)(2.20)}} \approx 8.05 \times 10^1\,\text{m s}^{-1}.$$

(d) Example: increase downforce coefficient c via aero package while controlling drag penalty on straights.

Q3

(a)

$$\Delta E_k = \frac{1}{2}m(u^2 - v^2) = \frac{1}{2}(7.98 \times 10^2)(92^2 - 28^2) \approx 3.06 \times 10^6\,\text{J}.$$

(b) Energy to discs: $0.80\Delta E_k \approx 2.45 \times 10^6\,\text{J}$. Per disc: $Q \approx 6.12 \times 10^5\,\text{J}$.

$$\Delta T = \frac{Q}{mc_d} = \frac{6.12 \times 10^5}{(1.20)(4.60 \times 10^2)} \approx 1.11 \times 10^3\,\text{K}.$$

(c)

$$P_{\text{brake,avg}} = \frac{0.80\Delta E_k}{2.6} \approx 9.4 \times 10^5\,\text{W}.$$

(d) Any two valid points: non-uniform disc heating, cooling during braking, energy partition to tyres/air, temperature-dependent c_d, pad/disc contact non-idealities.

Mark scheme (CIE 9702 P4 style, Variant 2)

Key

- **M**: method mark
- **A**: accuracy mark
- **B**: independent statement
- **C**: quality of explanation / evaluation

Q1 [14]

Part	Marking points	Marks
(a)	Uses $P = Fv$ and $F = ma$, obtains $a = P/(mv)$. (M1A1)	2
(b)	Correct differential setup (M1), integration (M1), use of $v = 0$ at $t = 0$ (A1), final expression correct (A1).	4
(c)	Correct substitution (M1), $t \approx 3.55\,\text{s}$ (A1).	2
(d)	Uses $P = kv^3$ at terminal (M1), rearranges and computes $v_t \approx 1.28 \times 10^2\,\text{m s}^{-1}$ (A1A1).	3
(e)	One quantitative + one physical justification, coherent comparison of models. (C1C1B1)	3

Q2 [13]

Part	Marking points	Marks
(a)	Identifies $N = mg + D$ (B1), states $F_{\text{max}} = \mu N$ (B1).	2
(b)	Equates mv^2/r to friction limit (M1), substitutes $D = cv^2$ (M1), rearranges to explicit v_{max} form (A1A1).	4
(c)	Correct substitution (M1), valid numerical result $\approx 8.0 \times 10^1\,\text{m s}^{-1}$ (A1A1).	3
(d)	Feasible change identified (B1), correct physics chain (C1), quantitative justification for $\geq 5\%$ potential (C1), clear evaluation of trade-off (C1).	4

Q3 [13]

Part	Marking points	Marks
(a)	Correct ΔE_k expression (M1), substitution (M1), $\Delta E_k \approx 3.06 \times 10^6\,\text{J}$ (A1).	3
(b)	Applies 80% transfer and four-disc split (M1M1), uses $Q = mc\Delta T$ (M1), obtains $\Delta T \approx 1.1 \times 10^3\,\text{K}$ (A1).	4
(c)	Uses $P = E/t$ with braking duration (M1), $\approx 9.4 \times 10^5\,\text{W}$ (A1).	2
(d)	Two distinct limitations with directional effect on ΔT explained. (C1C1B1B1)	4

Total: 40 marks